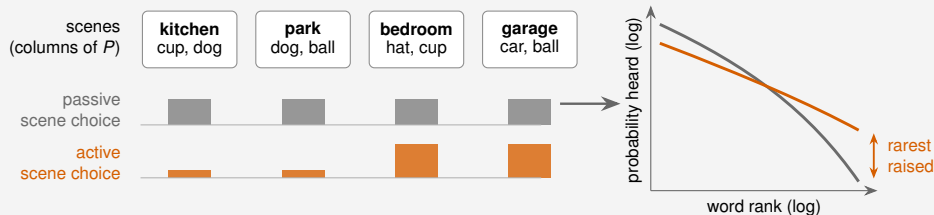


A Choosing scenes (active learning in the fast-mapping analysis)

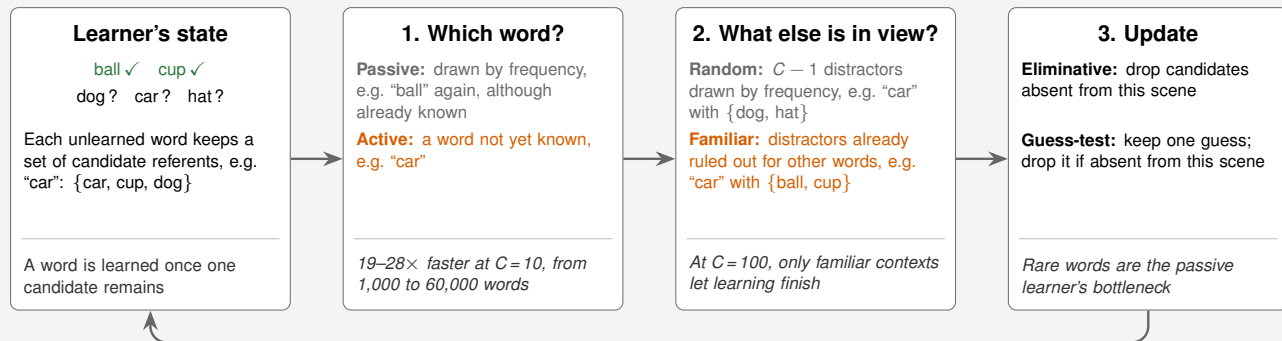


Passive: scenes at random, so the rarest words (car, hat) are heard very rarely.

Active: mix scenes to make the rarest word as likely as possible,
 $\hat{q} = \arg \max_q \min(Pq)$,
or estimate P online.

235× faster on SUN scenes, 6.7× on Flickr8k; online matches optimal.

B Composing an episode (incremental active learning)



*Under forgetting, returning to not-yet-learned words doubles as rehearsal:
the active/passive gap widens from 4.4× to 56× as memory gets less durable*